

# Quantum Mechanics II Exccercise 1

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## 1 Natrual Units

### 1.1 Unit Equivilence

In CGS, the speed of light is measured in  $cm/sec$ . Therefore, for a unit system where  $c = 1$ , we get:

$$c = 3 \cdot 10^{10} cm/sec = 1 \quad (1)$$

$$1 sec = 3 \cdot 10^{10} cm \quad (2)$$

Therefore, distance and time are measured using the same units. Using the same logic, in CGS, mass is measured in grams and energy is measured in ergs. Assigning equation (2) into the definition of 1 erg, we get:

$$1 erg = 1 g \cdot \frac{cm^2}{sec^2} = 1 g \cdot \frac{cm^2}{9 \cdot 10^{20} cm^2} = \frac{1}{9} \cdot 10^{-20} g \quad (3)$$

And so we got that, assuming  $c = 1$ , the units for mass and energy are unified. This can also be shown using the relation from special relativity  $E = mc^2$ . As instructed, the units of mass and energy will be  $eV$ .

Now let's use the fact that  $\hbar = 1$ .  $\hbar$  Has units of angular momentum. Angular momentum has units of  $energy \times time$ . Since  $\hbar$  is unitless, this means that angular momentum is unitless, and requires the units of energy to be opposite to the unit of time:

$$\hbar = 10^{-27} erg \cdot sec = 1 \quad (4)$$

$$1 sec = 10^{27} erg^{-1} \quad (5)$$

Therefore the units of distance and time are opposite to the unit of energy and mass, and  $eV^{-1}$  can be used to measure them. Writing all of the units in  $eV$  is now just an issue of conversion from erg.

$$1 erg = 6.242 \cdot 10^{11} eV \quad (6)$$

$$1 g = 9 \cdot 10^{20} erg = 5.617 \cdot 10^{32} eV \quad (7)$$

$$1 sec = 10^{27} erg^{-1} = 1.6 \cdot 10^{15} eV^{-1} \quad (8)$$

$$1 cm = sec/(3 \cdot 10^{10}) = 5.34 \cdot 10^4 eV^{-1} \quad (9)$$

### 1.2 Newton's Constant and Plank Mass

Using the values from equations (6) to (9):

$$G_N = 6.7 \cdot 10^{-8} cm^3 sec^{-2} g^{-1} \quad (10)$$

$$G_N = 6.7 \cdot 10^{-8} \cdot (5.34 \cdot 10^4 eV^{-1})^3 \cdot (1.6 \cdot 10^{15} eV^{-1})^{-2} \cdot (5.617 \cdot 10^{32} eV)^{-1} \quad (11)$$

$$G_N = 7.095 \cdot 10^{-57} eV^{-2} \quad (12)$$

This is the very close to the value we would get using Plank Mass  $M_{pl} = 1.2 \cdot 10^{19} GeV$ :

$$G_N = M_{pl}^{-2} = (1.2 \cdot 10^{19} GeV)^{-2} = 6.94 \cdot 10^{-57} eV^{-2} \quad (13)$$

With a difference of around 2.1%, resulting probably from our rounding of the numbers.